

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Examination – 2023

Course: B. Tech. Branch : E& TC Engineering Semester : V

Subject Code & Name: BTETOE505B Artificial Intelligence and Machine learning

Max Marks: 60

Date:

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
 3. Use of non-programmable scientific calculators is allowed.
 4. Assume suitable data wherever necessary and mention it clearly.
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(Level/CO) Marks

Q. 1 Solve any two of the followings.

- | | | |
|--|----------|----------|
| A) What is artificial intelligence? What are the types of environments in AI. | (L1/CO1) | 6 |
| B) Explain the agents and their types in AI. | (L1/CO2) | 6 |
| C) Explain in detail a state-space search with an example and turing test. | (L1/CO1) | 6 |

Q.2 Solve any two of the followings.

- | | | |
|---|----------|----------|
| A) Write a note on problem solving agents | (L2/CO2) | 6 |
| B) Write in detail about uninformed search strategies along with the advantages and disadvantages. | (L1/CO3) | 6 |
| C) Write and explain the A* algorithm with advantages and disadvantages. | (L1/CO1) | 6 |

Q. 3 Solve any two of the followings.

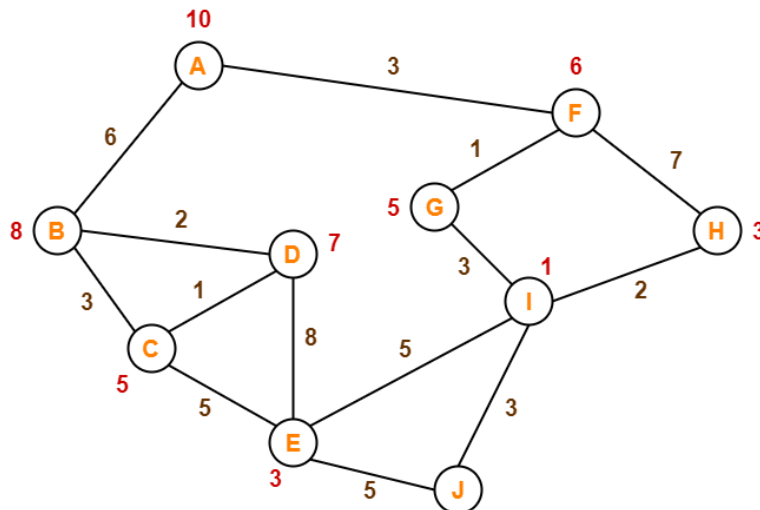
- | | | |
|--|----------|----------|
| A) Explain the Mini-Max algorithm along with the example. | (L2/CO2) | 6 |
| B) Write a note on cutting off search and forward pruning. | (L2/CO3) | 6 |
| C) Explain in detail the stochastic games and partially observable games along with the examples. | (L1/CO1) | 6 |

Q.4 Solve any two of the followings.

A) Consider the following graph-

(L1/CO2)

6

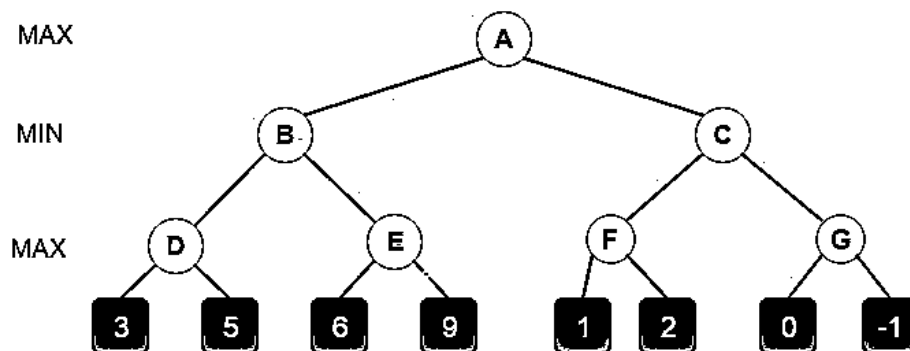


The numbers written on edges represent the distance between the nodes. The numbers written on nodes represent the heuristic value. Find the most cost-effective path to reach from start state A to final state J using A* Algorithm

B) Explain the alpha-beta search algorithm. Also solve the following example using Alpha beta pruning.

(L3/CO3)

6



C) Write a note on agents based on propositional logic.

(L1/CO1)

6

Q. 5 Solve any two of the followings.

A) Define classical planning. Explain the algorithms for planning as state-space search.

(L2/CO2)

6

B) What is ensemble learning? What are the types of ensemble learning?

(L2/CO3)

6

C) Comment on finding the hypothesis in supervised machine learning algorithm from given data with an example.

(L2/CO2)

6

***** End *****